University of Nevada, Reno

Optical Pumping of Rubidium Spin in a Solid Argon Matrix

A thesis submitted in partial fulfillment of the

requirements for the degree of Master of Science in Physics

by

Sameer K. Regmi

Dr. Jonathan D. Weinstein/Thesis Advisor

August 2013

©2013 - Sameer K. Regmi

All rights reserved.



THE GRADUATE SCHOOL

We recommend that the thesis
prepared under our supervision by

SAMEER K. REGMI

entitled

Optical Pumping of Rubidium Spin in a Solid Argon Matrix

be accepted in partial fulfillment of the
requirements for the degree of

MASTER OF SCIENCE

Jonathan D. Weinstein, Advisor

Andrew Geraci, Committee Member

Aleksey Telyakovskiy, Graduate School Representative

Marsha H. Read, Ph. D., Dean, Graduate School

August, 2013

Abstract

Using the matrix isolation technique we have trapped rubidium atoms inside a cryogenic solid argon crystal of thickness 200 μm. This weakly-interacting solid offers the experimental simplicity and high densities of solid-state systems, while allowing the implanted atoms to retain many of the properties of their gas-phase counterparts. As such, they are a promising environment for quantum information experiments, as well as sensors such as magnetometers.

We spectroscopically observed narrow lines at 799 nm, 785 nm, and 747 nm, and investigated optical pumping of the implanted rubidium atoms. We observed that the absorption spectrum of the rubidium could be modified by incident laser light. This modification could be reversed by applying an additional light source. In addition, we were able to create spin polarization and measure a longitudinal spin relaxation time T_1 of about 0.1 s.

Contents

Abstract i
Acknowledgements ii
Contents iii
List of Tables v
List of Figures vi

1 Introduction **1**
1.1 Atomic spin systems as qubits 1
1.2 Matrix isolated spin systems 3
1.2.1 Helium matrix isolation 4
1.2.2 Argon matrix isolation 6
1.3 Optical pumping 7

2 Apparatus **11**
2.1 Cryogenic parts 11
2.1.1 Cryostat 11
2.1.2 Oven, cover, and fill line 12
2.1.3 Horizontal plate, vertical plates, and sapphire window holder 16
2.1.4 Sapphire window 18
2.1.5 Photodetectors, spectrometer and cameras 19
2.2 Helmholtz coils 20
2.3 Chemicals 20
2.4 Data acquisition and analysis 21

3 Matrix Isolation **22**
3.1 Growing blank argon crystals 22
3.1.1 General procedure of argon deposition 23
3.1.2 Monitoring the crystal growth 24

3.1.3 Cleaning the sapphire window for next deposit 25
3.1.4 Growing argon crystal below 5 K 26
3.1.5 Growing argon crystal at 20 K 29
3.2 Growing argon crystal doped with rubidium 29
3.2.1 Growing Ar-Rb crystal at 15 K 34
3.2.2 Detecting rubidium in Ar-Rb gas beam 39
3.2.3 Power outage and its effect on spectrum 40
3.3 Rubidium density in the Ar-Rb crystal 41

4 Optical Pumping **43**
4.1 Problem of "bleaching" and its solution 43
4.1.1 Lamp as an unbleaching light source 45
4.1.2 LED as an unbleaching light source 48
4.2 Optical Pumping 50
4.2.1 Measurement of relaxation time T_1 51
4.2.2 Duration of pumping and its effect 58
4.2.3 Effect of temperature on the Ar-Rb crystal 62

5 Summary and Future Prospects **64**

A Mechanical Drawings **66**

B Rubidium Vapor Pressure Curve **72**

C Rubidium Hyperfine Structure **74**

Bibliography **79**

2.1.2), and a vacuum pump-manifold are measured by three vacuum transducer gauges from MKS Instruments [34]. The sensors can measure pressures as low as 1×10^{-5} torr but not below that limit. A residual gas analyzer RGA100 from Stanford Research Systems [35] is used to leak-check the cryostat with helium.

Inside the aluminum vacuum chamber there is an aluminum shielding box which encloses the parts needed for matrix isolation. This box wrapped with layers of aluminized mylar sheets helps to block the blackbody radiation due to room temperature. The outer chamber as well as the inner shielding box both have five windows each of diameter 3" and thickness 0.5" in order to have optical access. Cryogenic silicon diode sensors DT-670 [36], which are calibrated relative to [37], are attached to different parts of the cryostat in order to monitor temperature. Details about the design of the cryostat with cooling and vacuum system as well as windows and other accessories have been well explained in a previous dissertation [1].

To serve the purpose of our experiment of matrix isolation we modified the cryostat by removing an existing cell (which was used for buffer-gas cooling experiments in the past) and adding parts as described in the next subsections. Figure 2.1 shows the internal view of the cryostat along east-west direction, with the vacuum chamber and aluminum shielding box removed.

2.1.2 Oven, cover, and fill line

An oven and its cover were designed in order to contain rubidium. It is made of stainless steel and is cube-shaped, with length, width, and height of 1" each. Internally the oven has a cylindrical cavity of diameter 0.75" and depth also of 0.75". A cover is sealed with the oven using a teflon O-ring, and held tight with 4-40 UNC screws. The middle of the cover has a hole of diameter 0.124" which is used to connect the oven with the fill line. Figure 2.2 shows the oven with nozzle, fill line, a resistor heater,

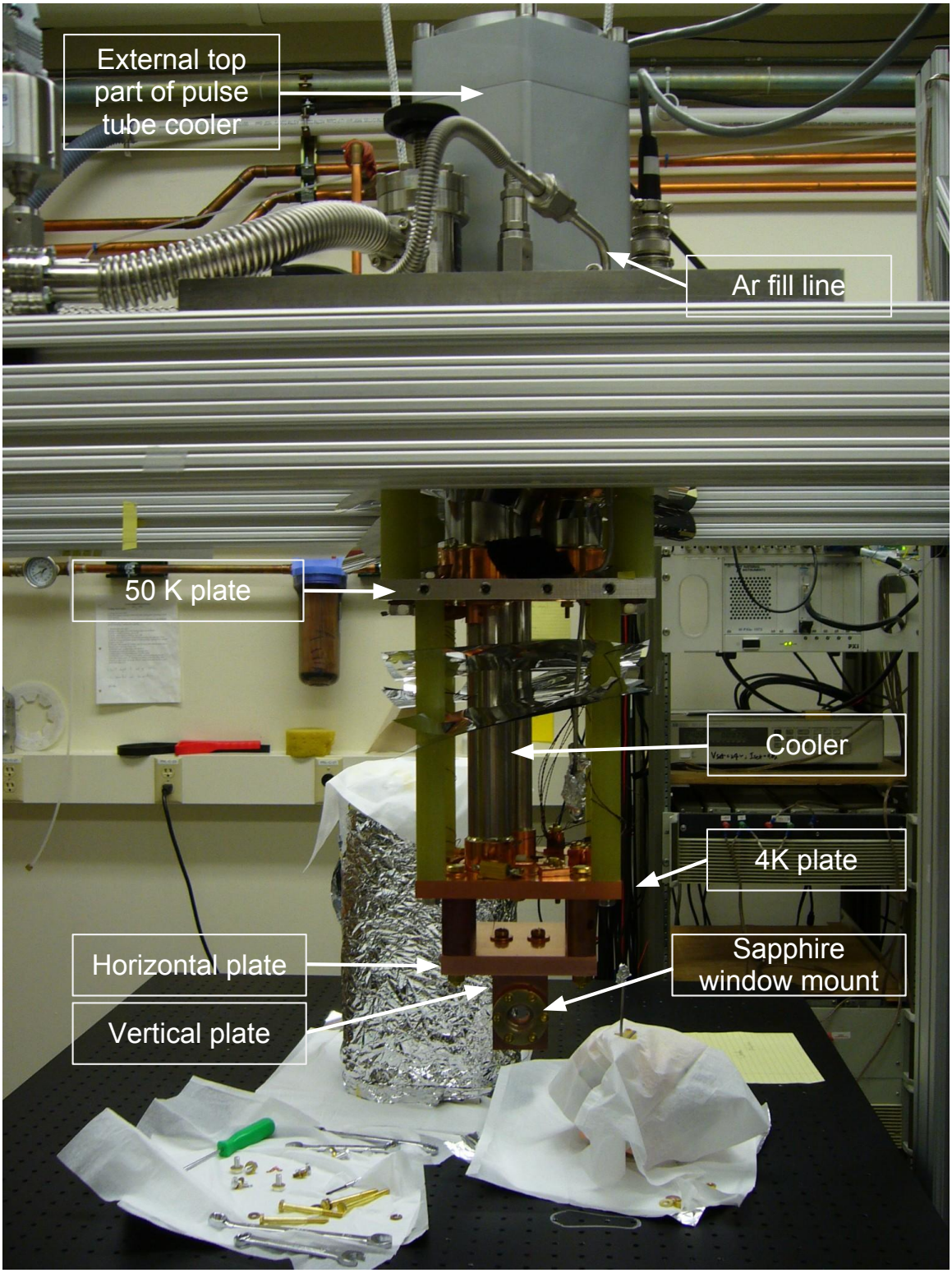


Figure 2.1: Internal view of the cryostat (oven not shown) along east-west direction, with the vacuum chamber and aluminum shielding box removed.

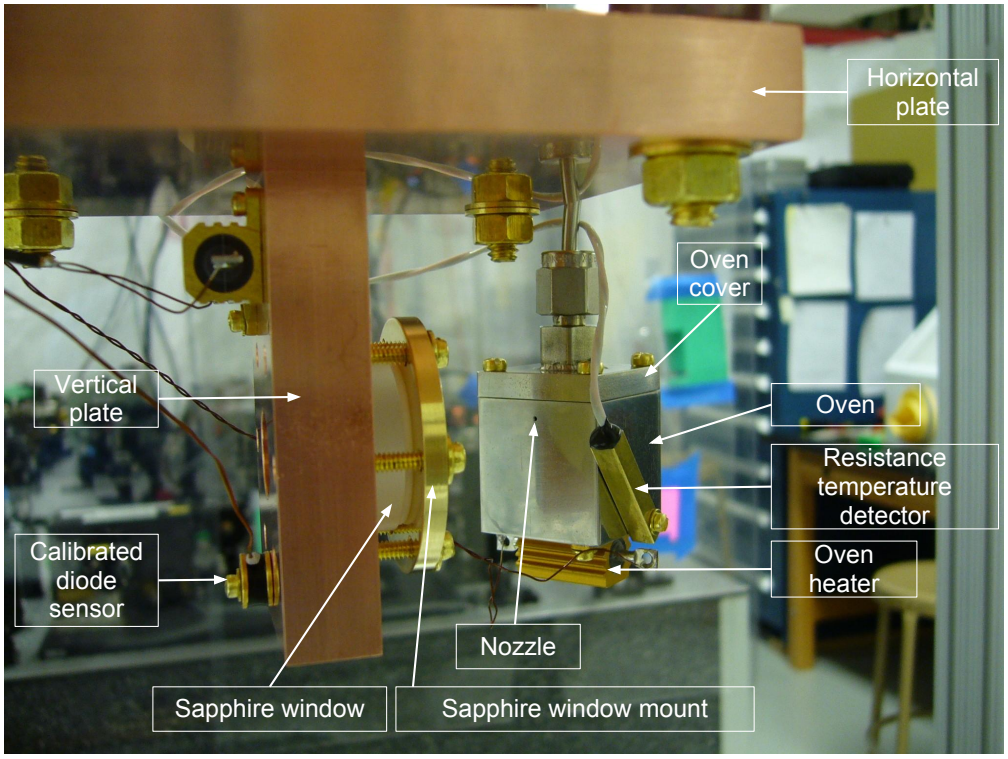


Figure 2.2: Oven with nozzle on the side facing the sapphire window.

and a temperature detector. Appendix A presents the mechanical drawings of the oven and cover.

A resistor heater of 100 Ω [38], which can be attached to the base or a side of the oven, is used to heat the rubidium sample contained inside the oven so that the amount of rubidium deposited can be controlled by adjusting the temperature. In order to monitor the oven temperature a flat platinum resistance detector [39] is attached to one side of the oven.

The side of the oven facing the sapphire window has a clear hole of diameter 0.020". This hole acts as a nozzle for the flow of argon with rubidium during deposition. As can be seen in figure 2.3, the oven is placed slightly right of the perpendicular axis of sapphire window for the purpose of clear optical access to the window. The nozzle side of oven is making an angle $15 - 20^\circ$ with the plane of the sapphire window. The

distance between the nozzle and the sapphire window is about 2".

To control the right amount of flow of rubidium and argon gas during formation of an Ar-Rb crystal, we need to design the correct diameter of the nozzle of the oven. Below we present the calculation for the nozzle diameter.

We assume that 100 μm thick Ar-Rb crystal can be formed in 1 hour and that the ratio of number of rubidium atoms to argon atoms will be 1 to 1000. Then the ratio of argon pressure (P_{Rb}) to rubidium pressure (P_{Ar}) should be $\frac{P_{Rb}}{P_{Ar}} = \frac{1}{1000}$. If we assume that $P_{Ar} = 10$ torr, then we have $P_{Rb} = 1 \times 10^{-2}$ torr. From the rubidium vapor pressure curve in appendix B we see that the required oven temperature to get 1×10^{-2} torr of rubidium pressure is $T_{oven} = 440$ K.

Next, we find the throughput of argon to deposit 100 μm thick Ar-Rb crystal on the sapphire window. We assume that during deposition the Ar-Rb beam will be sprayed into 2π steradians. Then at a distance of 2", the total volume of the argon solidified would be 1.6×10^{12} μm^3. The density of solid argon is 1.768 g cm^{-3} at 20 K [40], and therefore the mass of solidified argon is 2.9 g. With argon atomic weight 40 and Avogadro's number 6.022×10^{23} we find that the number of argon atoms solidified is $N_{Ar} = 4.3 \times 10^{22}$. Then using the ideal gas law we find $P_{Ar} \cdot V_{oven} = N_{Ar} \cdot K \cdot T_{oven} = 2.0 \times 10^3$ torr liters at 440 K. Then assuming that argon deposition rate is 100 μm per hour, we find the throughput of argon to be $Q_{Ar} = 0.56$ torr liter/sec.

Next, we find the conductance of the oven nozzle, C_n. The throughput is given by [41]

$$Q_{Ar} = (P_{Ar} - P_n)C_n \qquad (P_{Ar} > P_n), \tag{2.1}$$

where P_n is the pressure at the end of nozzle which we assume to be zero. From this we see that $C_n = \frac{Q_{Ar}}{P_{Ar}} = 5.6 \times 10^{-2}$ liter/sec.

The mean free path of argon, at the pressure of 10 torr and at room temperature,

is $\sim$5 μm [41], which is much smaller than the nozzle diameter, and therefore the flow of gas in our system is viscous rather than molecular. We want the viscous flow to be in laminar and not turbulent regime. In laminar regime, at 20 °C, the conductance of a long tube having a circular cross section is given by [41, 42]

$$C_n = 180\frac{{D_n}^4}{L_n}P_{Ar} \text{ liter sec}^{-1} \tag{2.2}$$

where D_n is the diameter (in cm) and L_n is the length of the nozzle (in cm), P_{Ar} is the average argon pressure in oven (in torr).

Then with $L_n = 0.318$ cm, $C_n = 5.6 \times 10^{-2}$ liter/sec, and $P_{Ar} = 10$ torr, equation 2.2 gives us $D_n = 0.06$ cm (0.02") for the nozzle. Therefore, we have designed the oven with nozzle having diameter of 0.020" with tolerance $\pm$0.005 as shown in figure A.1 of appendix A.

A pressure gauge is attached to the fill line and we use the pressure there (P_g) to get information about oven pressure(P_{Ar}), assuming $P_g \approx P_{Ar}$. This assumption will be true when $C_n \ll C_f$, where (C_f) is the conductance of the fill line. The fill line has an inner diameter of $\sim$0.3 cm and length of about 50 cm. The nozzle has the diameter of 0.06 cm and length of 0.125 cm. Then using these values in equation 2.2 we see that $\frac{C_f}{C_n} = 2\frac{P_g+P_{Ar}}{P_{Ar}} \geq 4$. This suggests that our measurement of oven pressure could be off by at most 25%.

2.1.3 Horizontal plate, vertical plates, and sapphire window holder

In order to hold the sapphire window we designed a horizontal plate, a vertical plate, and a holder which are shown in figure 2.3. Appendix A shows their mechanical drawings.

The horizontal plate is a square shaped slab made of copper (110) having length